

LC 1105: Bookcase Shelf Filling

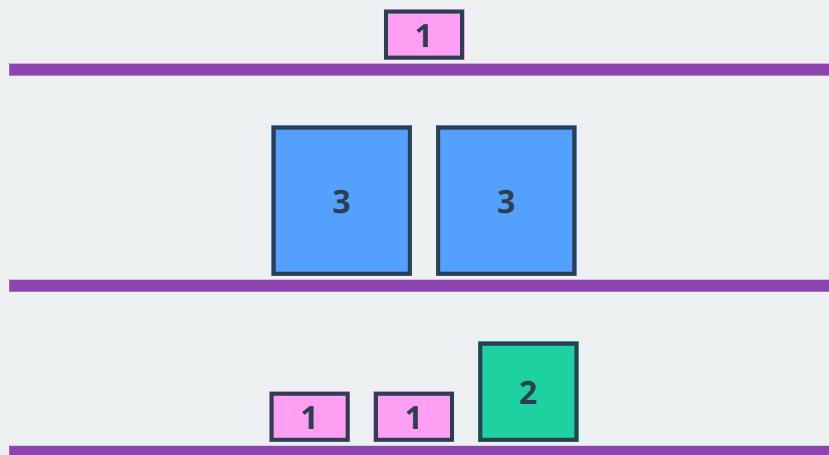
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PROBLEM UNDERSTANDING

- each book have → thickness and height
- but shelf width is limited
- need to output min possible height of the stack, can place rows of book over other rows

Eg: numbers written inside represent height



$$1 + 3 + 2 = 6m \text{ height}$$

Constraint → need to place in order in which books are given. first top row is filled

SOLUTION APPROACH

hence it's a partition problem as need to decide when to stop adding books to a row for min height

- number of parts not given, so simply $dp[i][k] \rightarrow dp[i] = \text{min total height needed to place first } i \text{ books}$
- height of a row is $\text{max}(\text{height of books in that row})$

CODE IMPLEMENTATION (C++)

```
class Solution {
public:
    int minHeightShelves(vector<vector<int>>& books, int shelfwidth) {

        int n=books.size();
        vector<int> dp(n+1,INT_MAX);
        dp[0]=0;

        //dp[i]=minheight to place first i books

        for(int i=1;i<=n;i++){

            int w=0;
            int h=0;
            for(int j=i-1;j>=0;j--){

                w+=books[j][0];
                if(w>shelfwidth) break;

                h=max(h,books[j][1]);
                int curr=dp[j]+h;

                dp[i]=min(dp[i],curr);

            }
        }
        return dp[n];
    }
};
```

